

Selective Activation of the JNK Signaling Cascade and c-Jun Transcriptional Activity by the Small GTPases Rac and Cdc42Hs

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Summary

The Rho subfamily of GTPases is involved in control of cell morphology in mammals and yeast. The mammalian Rac and Cdc42 proteins control formation of lamellipodia and filopodia, respectively. These proteins also activate MAP kinase (MAPK) cascades that regulate gene expression. Constitutively activated forms of Rac and Cdc42Hs are efficient activators of a cascade leading to JNK and p38/Mpk2 activation. RhoA did not exhibit this activity, and none of the proteins activated the ERK subgroup of MAPKs. JNK, but not ERK, activation was also observed in response to Dbl, an oncoprotein that acts as a nucleotide exchange factor for Cdc42Hs. Results with dominant interfering alleles place Rac1 as an intermediate between Ha-Ras and MEK1 in the signaling cascade leading from growth factor receptors and v-Src to JNK activation. JNK and p38 activation are likely to contribute to the biological effects of Rac, Cdc42Hs, and Dbl on cell growth and proliferation.

Introduction

Members of the Ras superfamily of small GTP-binding proteins have important regulatory roles in diverse cellular functions, including mitogenesis, cytoskeletal rearrangements, transcytosis, and motility (reviewed by Bokoch and Der, 1993; Bourne et al., 1990; Downward, 1990; Hall, 1993). Whereas the Rab and Arf subfamilies control vesicular protein trafficking (reviewed by Ferro-Novick and Jehn, 1994; Rothman, 1994), the Ras and Rho subfamilies function in signal transduction. The Ras proteins link growth factor receptor activation to gene regulation through mitogen-activated protein kinase (MAPK) cascades that eventually lead to phosphorylation of specific transcription factors (reviewed by Marshall, 1995). Growth factor receptor activation results in recruitment of the nucleotide exchange factor Sos to the plasma membrane, where it activates Ras by catalyzing the exchange of GDP by GTP (Aronheim et al., 1994; Buday and Downward, 1993). The Ras-GTP form recruits the serine/threonine kinase Raf to the plasma membrane, thereby permitting its activation by signals yet to be identified (Leevers et al., 1994; Stokoe et al., 1994). Once activated, Raf phosphor-

ylates and activates the dual specificity kinases MEK1 and MEK2, whose substrates are the extracellular signal-regulated ERK1 and ERK2 MAPKs (reviewed by Cobb and Goldsmith, 1995; Marshall, 1995). The activated ERKs phosphorylate several important cytoplasmic substrates and also translocate to the nucleus, where they phosphorylate transcription factors leading to immediate-early gene induction (reviewed by Marshall, 1995; Hill and Treisman, 1995). The use of constitutively activated and catalytically inactive MEK1 mutants has offered the view that activation of the ERK cascade is both necessary and sufficient for oncogenic transformation and presumably for a mitogenic response (Cowley et al., 1994; Mansour et al., 1994). Indeed, an effector domain mutant of oncogenic Ha-Ras(V12) that cannot bind Raf-1 and activate the ERK cascade, Ha-Ras(V12;E37G), is nontransforming (White et al., 1995). However, another effector domain mutant, Ha-Ras(V12;T35S), which is fully capable of binding Raf and stimulating ERK activity, is compromised in its transforming activity, suggesting that additional Ha-Ras effectors contribute to cell transformation and presumably mitogenesis (White et al., 1995). The identity of these effectors is not clear.

Recently, downstream mediators linking Ras activation to nuclear events were identified. Ras activation was shown to play a critical role in activation of members of the MAPK group, the c-Jun N-terminal kinases (JNKs) (also known as stress-activated protein kinases [SAPKs]; Dérijard et al., 1994; Kyriakis et al., 1994), in response to growth factors (Minden et al., 1994). Like the ERKs, the activated JNKs translocate to the nucleus (F. Dolfi and M. K., unpublished data), where they phosphorylate transcription factors such as c-Jun (Hibi et al., 1993; Su et al., 1994) and ATF2 (Gupta et al., 1995). Although Ha-Ras function is essential for JNK activation by growth factors (but not by cytokines, such as tumor necrosis factor α [TNF α]), oncogenically activated Ha-Ras leads only to partial JNK activation (Minden et al., 1994). By contrast, Ha-Ras is sufficient for full ERK activation (de Vries-Smits et al., 1992; Robbins et al., 1992; Thomas et al., 1992; Wood et al., 1992). Another difference between the JNK and ERK cascades is that Raf is not directly involved in JNK activation (Minden et al., 1994). Instead, the JNK cascade is strongly activated by the MAPKKK MEK1, which, unless overexpressed, does not lead to ERK activation (Minden et al., 1994; Yan et al., 1994). Although MEK1 activity is stimulated by growth factors in a Ras-dependent manner (Lange-Carter et al., 1993; Lange-Carter and Johnson, 1994), the mechanism of its activation is not clear. In yeast, activation of an MEK1 homolog, STE11, by the pheromone receptor appears to involve activation of another protein kinase, STE20 (reviewed by Herskowitz, 1995).

Members of the Rho subfamily, although less extensively studied than the Ras subfamily, also play critical roles in growth factor signaling through regulation of the actin cytoskeleton (reviewed by Ridley, 1994). Rac regulates membrane ruffles (lamellipodia) through formation

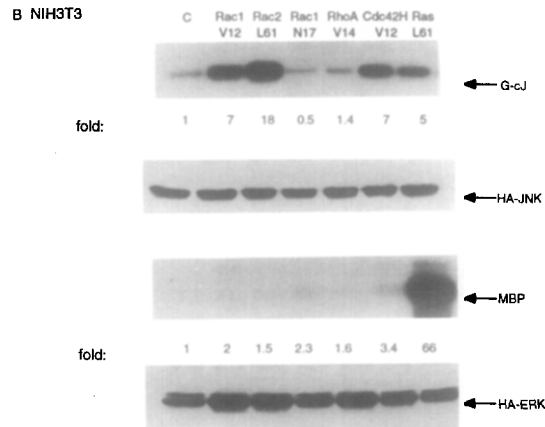
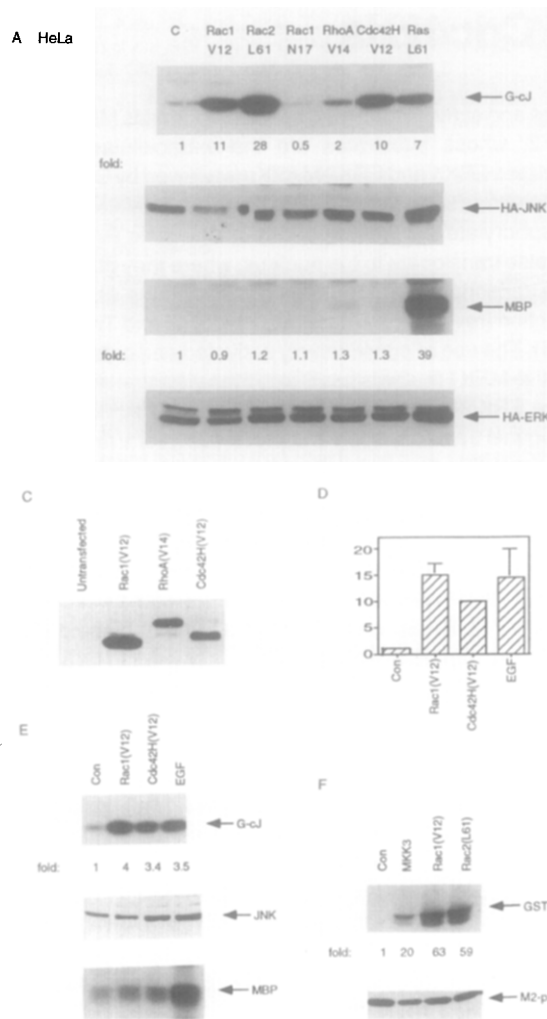


Figure 1. Activation of JNK and p38 but Not ERK by Activated Rac and Cdc42Hs

(A) HeLa cells were transfected with 1 μ g of either pSR α -HA-JNK1 or pSR α -HA-ERK2 together with 1 μ g of either empty pSR α expression vector (lane C) or the indicated small GTPase expression vectors. JNK and ERK activities were measured by immunocomplex kinase assays using GST-c-Jun (G-cJ) or MBP as substrates, respectively. A sample of each lysate was analyzed for expression of HA-JNK or HA-ERK by Western blot analysis using anti-HA antibody. The segments of the autoradiograms and fluorograms that contain the phosphorylated substrates and the expressed kinases are shown. The fold stimulation of kinase activity as quantitated with a phosphorimager is indicated below each lane.

(B) NIH 3T3 cells were transfected as described in (A) using 0.5 μ g of each of the indicated small GTPase expression vectors or 0.2 μ g of Cdc42Hs(V12) expression vector. Total DNA concentration was kept constant using pSR α . Kinase activities were determined as described above. These experiments were repeated at least three times with similar results.

(C) HeLa cell lysates from (A) were analyzed by immunoblotting for expression of Myc-tagged Rac1(V12), RhoA(V14), and Cdc42Hs(V12). (D) HeLa cells were transfected with 1 μ g of pSR α -HA-JNK1 together with either empty expression vector (lane C), Rac1(V12), or Cdc42Hs(V12) expression vectors. Cells transfected with HA-JNK1

and empty vector were treated with or without EGF (15 ng/ml) for 15 min. JNK activity was measured as described in (A). The fold stimulation over unstimulated cells, as quantitated by phosphorimager, is presented. These experiments were repeated three times with similar results.

(E) Rat1A cells stably expressing Rac1(V12) or Cdc42Hs(V12) and parental cells (Con) were assayed for endogenous JNK or ERK activity by immunocomplex kinase assays with GST-c-Jun (G-cJ) or MBP, respectively, as substrates. Parental cells were treated with or without EGF (15 ng/ml for 10 min). JNK1 expression level was measured by immunoblotting. These experiments were repeated twice with similar results.

(F) HeLa cells were transfected with pCMV5-M2-p38 (1 μ g) together with empty expression vector (Con) or the indicated expression vectors. p38 activity was determined by immunocomplex kinase assay using GST-ATF2 Δ as substrate. A sample of each immunoprecipitate was analyzed for its content of M2-p38 by immunoblotting. This experiment was repeated twice with similar results.

of cortical actin fibers (Ridley et al., 1992), whereas Rho controls formation of stress fibers (Ridley and Hall, 1992). Rac is also involved in activation of the NADPH oxidase complex in neutrophils (reviewed by Segal and Abo, 1993). Recently, another member of this subfamily, human Cdc42 (Cdc42Hs), was found to be involved in filopodia formation (Kozma et al., 1995; Nobes and Hall, 1995). Like the Ras proteins, the Rho-like proteins are believed to be activated in response to growth factors and polypeptide hormones (Hall, 1993; Nobes and Hall, 1994; Ridley, 1994). With the exception of NADPH oxidase activation, the Rho-like proteins were hitherto thought to function exclusively in the regulation of cell morphology, an evolutionary conserved role that is also displayed by *Saccharomyces cerevisiae* Cdc42 (Cdc42Sc), which regulates bud site selection and cell polarity (Adams et al., 1990). Both

Rac1 and Cdc42Hs in their GTP-loaded states can bind to the p21-activated serine/threonine kinase PAK65 and stimulate its autophosphorylation activity (Manser et al., 1994; Martin et al., 1995). Although PAK65 is a STE20 homolog, its substrates or biological functions remain unknown. Recently, Rac1 was found to be an important mediator of oncogenic transformation by Ha-Ras and capable of cooperating with a membrane-targeted form of Raf (Raf-CAAX; Qiu et al., 1995). Here we describe a role for Rac and Cdc42Hs in signal transduction to the nucleus. Constitutively activated Rac1/Rac2 and Cdc42Hs, but not RhoA, are efficient and specific activators of the JNK cascade, leading to increased c-Jun transcriptional activity. These proteins are also efficient activators of another MAPK, p38/Mpk2 (Han et al., 1994; Lee et al., 1994; Rouse et al., 1994), but they do not stimulate ERK activity. A dominant

negative Rac1 mutant, Rac1(N17), attenuated JNK activation by growth factors, v-Src, and Ha-Ras. We propose that Rac1 and (possibly) Cdc42Hs are important intermediates to the JNK and p38 activation cascades and serve to coordinate inputs from growth factor receptors and other extracellular stimuli.

Results

Activation of JNK and p38 but Not ERK by Activated Rac Proteins

To examine whether other members of the Ras superfamily, in addition to Ha-Ras, can stimulate mammalian MAPK cascades, we used expression vectors encoding constitutively activated versions of Rac1, Rac2, RhoA, and Cdc42Hs. These GTPases were activated either by substitution of G12 or its equivalent (G14 in RhoA) by a valine residue (Ridley et al., 1992; Ridley and Hall, 1992; Kozma et al., 1995) or Q61 by a leucine residue (Xu et al., 1994). To test the effects on JNK and ERK activity, we cotransfected the different GTPase expression vectors with plasmids encoding hemagglutinin (HA) epitope-tagged versions of JNK1 and ERK2 (Minden et al., 1994). After 48 hr, the transiently expressed HA-JNK1 and HA-ERK were isolated by immunoprecipitation, their activities were determined by phosphorylation of either glutathione S-transferase (GST)-c-Jun(1-79) or myelin basic protein (MBP), respectively, and their expression levels were determined by immunoblotting. Rac1, Rac2, and Cdc42Hs were efficient JNK1 activators in both HeLa (Figure 1A) and NIH 3T3 (Figure 1B) cells. In both cell types, Rac and Cdc42Hs were more effective JNK activators than Ha-Ras(L61). In contrast, RhoA(V14) did not stimulate JNK activity, despite its efficient expression (Figure 1C). None of the Rho-like proteins had a considerable effect on ERK2 activity, which was strongly stimulated by Ha-Ras(L61) (Figures 1A and 1B). Rac1(N17), which, like Ha-Ras(N17), is inactive owing to its higher affinity toward GDP (Feig and Cooper, 1988; Ridley et al., 1992), did not activate JNK and, in fact, reduced its basal activity. The stimulation of JNK activity by either Rac1(V12) or Cdc42Hs(V12) was as efficient as its stimulation by either epidermal growth factor (EGF) (Figure 1D) or TNF α (data not shown; see, however, Figure 4) treatment of the same cells. Titration experiments indicated that near-maximal JNK activation was observed with either 50 or 200 ng of the Cdc42Hs(V12) or Rac1(V12) expression vectors, respectively. At these levels, their expression did not exceed the endogenous level of their wild-type counterparts by more than 6-fold (Cdc42Hs) or 1.5-fold (Rac1).

We measured JNK activity (both JNK1 and JNK2) in Rat1A cells stably expressing Rac1(V12) or Cdc42Hs(V12). In both cases, it was elevated in comparison with the unstimulated parental cells (Figure 1E). The extent of JNK activation was similar to its response to EGF, which, owing to a higher basal JNK activity in these cells, was lower than the usual 10- to 20-fold seen in other cell lines. Even in these stable transformants, Rac1(V12) and Cdc42Hs(V12) did not stimulate ERK activity. Immunoblot analysis indicated that the levels of Rac1(V12) and Cdc42Hs(V12) ex-

pression in these stably transfected clones were no more than 2- to 3-fold higher than the levels of their endogenous wild-type counterparts (data not shown). As shown below, Rac and Cdc42Hs also stimulate JNK activity in COS-1 cells (see Figure 7).

Recently, a member of the MEK subfamily, SEK1/MKK4/JNKK, was identified as an activator of both the JNKs and p38/Mpk2 (Dérjard et al., 1995; Lin et al., 1995). Another member of the MEK group, MKK3, is a specific p38 activator (Dérjard et al., 1995). To examine the effect of Rac1(V12) and Rac2(L61) on p38 activity, we cotransfected them with an M2-p38 expression vector. Immuno-complex kinase assays using GST-ATF2 as a substrate indicated that both Rac1 and Rac2 were potent p38 activators, leading to more efficient activation than MKK3 (Figure 1F).

Functional Relationships among the Small G Proteins

Ha-Ras(L61)-induced ruffling is inhibited by Rac1(N17) (Ridley et al., 1992). To examine whether Rac1(N17) also exhibits a dominant negative effect on JNK activation by Ha-Ras, we cotransfected the HA-JNK1 and Ha-Ras(L61) expression vector with or without the Rac1(N17) vector. Expression of Rac1(N17) led to almost complete inhibition of JNK activation by Ha-Ras(L61) (Figure 2A). No effect of Rac1(N17) on Ha-Ras (L61) expression was detected (data not shown). On the other hand, Ha-Ras(N17) in amounts sufficient for nearly complete inhibition of JNK activation by EGF (Minden et al., 1994; see below) had only a minor effect on JNK activation by Rac2(L61) (Figure 2A). These results suggest that Rac acts downstream of Ha-Ras in the pathway leading to JNK activation.

Next, we examined the relationships between Rac1/Rac2 and Cdc42Hs. We found that Rac1(N17) partially inhibited the activation of HA-JNK1 by the cotransfected Cdc42Hs(V12) expression vector (Figure 2B), but not by Rac1(V12) (data not shown). Cdc42Hs(N17) partially inhibited HA-JNK1 activation by cotransfected Rac2(L61), but somewhat less efficiently than Rac1(N17) inhibited the stimulation by Cdc42Hs(V12). HA-JNK1 activation by Cdc42Hs(V12) was completely inhibited by cotransfection of a truncation mutant of human PAK65. This mutant, PAKR(1-225), contains the regulatory domain that binds to either Rac1·GTP or Cdc42Hs·GTP but lacks the catalytic domain (Martin et al., 1995; A. A., unpublished data). Cotransfection of a RhoA(N19) mutant had only a marginal effect on JNK activation by Rac2(L61) (Figure 2B). Thus, while RhoA is not involved in JNK activation, we were not able to draw clear conclusions whether Cdc42Hs acts upstream of Rac1 and Rac2, as previously suggested by examination of their morphological effects (Nobes and Hall, 1995; Kozma et al., 1995). However, the effect of Cdc42Hs(N17) on Rac1-induced lamellipodia formation has not been examined.

Although Ha-Ras activity is necessary for JNK activation in response to growth factors, Ha-Ras(L61) is considerably less effective than EGF in activating JNK (Minden et al., 1994). We therefore examined whether Ha-Ras(L61) and Rac1(V12) can act additively to stimulate JNK activity. We

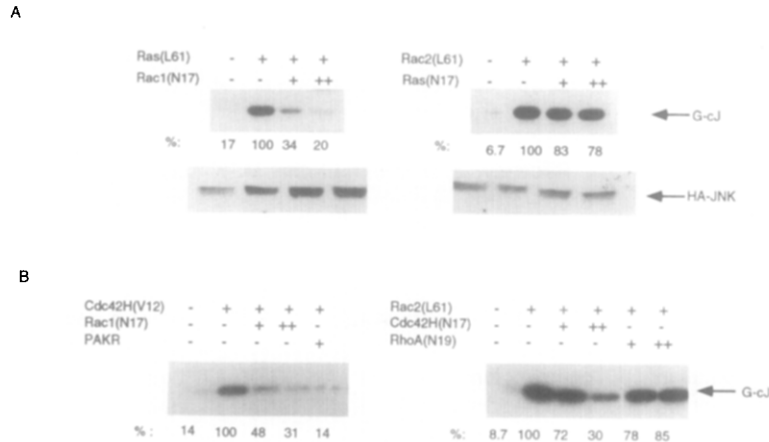


Figure 2. Relationships among Small GTPases Involved in JNK Activation (A) HeLa cells were transfected with 0.1 μ g of pSR α -HA-JNK1 and 1 μ g of pSR α -Ha-Ras(L61) in the absence or presence of either 0.5 or 1 μ g of pEXV-Rac1(N17) (left) or with 1 μ g of pSR α -Rac2(L61) in the absence or presence of 0.1 or 0.4 μ g of pSR α -Ha-Ras(N17) (right). Even the lower dose of Ha-Ras(N17) is sufficient to block completely growth factor activation of JNK (data not shown). JNK activity was measured by immunocomplex kinase assays, and HA-JNK expression was measured by immunoblotting. After quantitation with a phosphorimager, the highest level of JNK activity was given an arbitrary value of 100%, and the other activities are shown relative to this value. G-cJ, GST-c-Jun. (B) HeLa cells were transfected with 0.1 μ g of pSR α -HA-JNK1 and 1 μ g of Rac2(L61) or Cdc42HS(V12) in the absence or presence of 0.5 and 1 μ g of Cdc42HS(N17), or 0.5 and 1 μ g of RhoA(N19) expression vectors, as indicated. HA-JNK1 activity was determined and calculated as described above. All of these experiments were repeated at least twice with similar results.

found that cotransfection of the HA-JNK1 expression vector with a saturating amount of a Ha-Ras(L61) plasmid and a suboptimal amount of a Rac1(V12) expression vector resulted in very efficient JNK activation that exceeded the sum of the two individual responses (Figure 3). Wild-type Rac1 did not potentiate the response to Ha-Ras(L61) (data not shown).

Rac1 Acts Downstream of Tyrosine Kinases

We examined the effects of Rac1(N17) on JNK activation by either EGF or TNF α . Cotransfection with the Rac1(N17) plasmid partially inhibited the activation of HA-JNK1 by EGF, but hardly affected its response to TNF α (Figure 4).

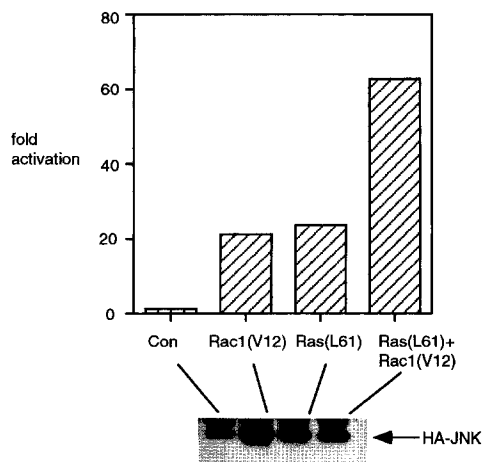


Figure 3. Coexpression of Rac1(V12) and Ha-Ras(L61) Leads to Strong JNK Activation

NIH 3T3 cells were transfected with 0.1 μ g of pSR α -HA-JNK1 and either 1 μ g of empty vector (Con), 0.2 μ g of Rac1(V12), or 1 μ g of Ha-Ras(L61) expression vectors (alone or together), as indicated. HA-JNK1 activity was measured by immunocomplex kinase assays, and the level obtained by cotransfection with the empty expression vector was given an arbitrary value of 1. HA-JNK1 expression was analyzed by immunoblotting and is shown at the bottom.

Nearly complete inhibition of the response to EGF was obtained by cotransfection of the Ha-Ras(N17) expression vector that, as previously observed (Minden et al., 1994), did not inhibit JNK activation by TNF α .

T cell receptor (TCR) activation stimulates JNK activity (Su et al., 1994). Unlike the EGF receptor, the TCR has no intrinsic kinase activity, and instead it operates through activation of nonreceptor tyrosine kinases (Weiss and Littman, 1994). We therefore examined JNK activation in response to a typical nonreceptor tyrosine kinase, v-Src. Cotransfection with a v-Src expression vector resulted in efficient JNK activation (Figure 5A). Considerable inhibition was obtained by cotransfection with Rac1(N17), and

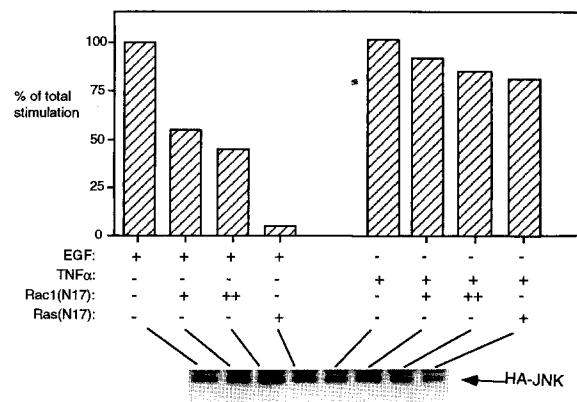


Figure 4. Dominant Negative Rac1 Inhibits JNK Activation by EGF

HeLa cells were transfected with 1.0 μ g of HA-JNK1 and either empty expression vector or 0.5 or 1 μ g of Rac1(N17) or 0.5 μ g of Ha-Ras(N17) expression vector. After 48 hr, cells were treated either with EGF (15 ng/ml) or human TNF α (10 ng/ml) for 15 min prior to harvesting. HA-JNK1 activity was measured by immunocomplex kinase assays, and expression level was determined by immunoblotting. In each set of experiments, the kinase activities are expressed relative to those obtained in cells cotransfected with the empty vectors, which were given arbitrary values of 100% (EGF, 15-fold stimulation; TNF α , 10-fold stimulation). The experiment was repeated three times with similar results.

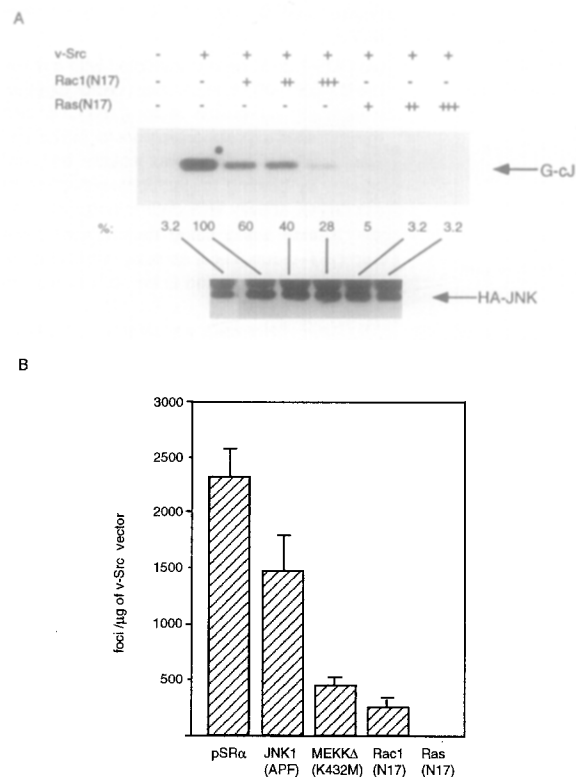


Figure 5. Rac1(N17) Inhibits JNK Activation and Foci Formation by v-Src

(A) NIH 3T3 cells were transfected with 0.1 μ g of pSR α -HA-JNK1 and 2 μ g of empty expression vector or 1 μ g of v-Src expression vector either alone or with 0.5, 1, or 2 μ g of either Rac1(N17) or Ha-Ras(N17) plasmids, as indicated. HA-JNK1 activity was measured by immunocomplex kinase assay, and its expression was examined by immunoblotting. JNK activity is expressed relative to the maximal level seen in v-Src-transfected cells, which represents 31-fold stimulation over the activity found in cells cotransfected with the empty vector. G-cJ, GST-c-Jun.

(B) NIH 3T3 cells were transfected with 0.1 μ g of v-Src expression vector in the presence of 2 μ g of the indicated expression vectors and carrier DNA. After 14 days, the cells were stained, and the foci were counted. The data from three experiments were averaged and are presented as foci per microgram of v-Src expression vector.

complete inhibition was achieved by cotransfection with Ha-Ras(N17). Interestingly, the dominant negative Rac1 and Ha-Ras mutants had similar relative effects on v-Src-mediated foci formation (Figure 5B). While cotransfection with Rac1(N17) inhibited focus formation by v-Src by 89%, cotransfection with Ha-Ras(N17) resulted in complete inhibition. Partial inhibition of v-Src-mediated transformation was also achieved by cotransfection of expression vectors encoding catalytically inactive forms of MEKK1 (81% inhibition) and JNK1 (36% inhibition).

JNK Activation by the Dbl Oncogene Product

Members of the Rho subfamily are activated by a variety of exchange factors that contain Dbl homology domains (reviewed by Ridley, 1994). It remains to be determined which exchange factor is specific to each of the Rho-like proteins in vivo. We examined the effects of the Dbl onco-

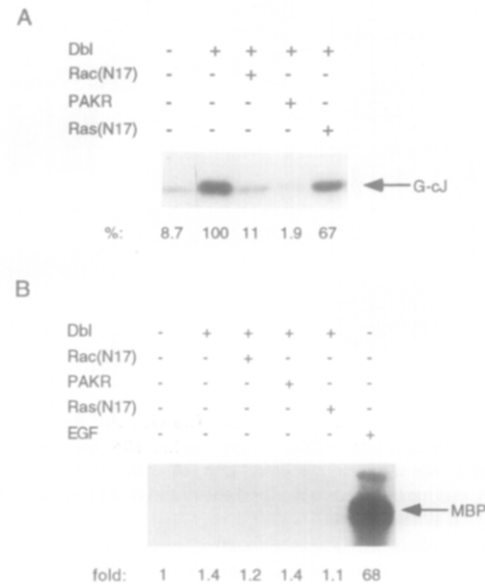


Figure 6. Rac1(N17) and PAKR Block JNK Activation by the Dbl Oncoprotein

(A) HeLa cells were transfected with 0.1 μ g of pSR α -HA-JNK1 and either 2 μ g of empty vector, 1 μ g of Dbl, 1 μ g of Rac1(N17), 1 μ g of PAKR, or 0.5 μ g of Ras(N17) expression vectors, as indicated. JNK activity was measured by immunocomplex kinase assay and is expressed relative to the maximal level seen in Dbl-transfected cells (100% equals 11.5-fold stimulation). G-cJ, GST-c-Jun.

(B) HeLa cells were transfected as described above, except that HA-ERK2 was used instead of HA-JNK1. Cells transfected with HA-ERK2 and empty vector were treated with or without EGF (15 ng/ml for 15 min). ERK activity was measured by immunocomplex kinase assays using MBP as substrate.

protein (Eva and Aaronson, 1985) on JNK activity. In vitro, Dbl acts as an exchange factor for Cdc42Hs and RhoA, but not Rac1 (Hart et al., 1991, 1994). Cotransfection with a Dbl expression vector resulted in efficient JNK activation (Figure 6A). To determine whether the response to Dbl is mediated through Cdc42Hs or RhoA, we used the PAKR construct, since PAK65 binds Rac1 and Cdc42Hs but not RhoA (Manser et al., 1994; Martin et al., 1995). Cotransfection with either Rac1(N17) or PAKR inhibited JNK activation by Dbl, but Ha-Ras(N17) had only a small effect on this response. Neither Rac1(N17) nor PAKR inhibited Dbl expression, as determined by immunoblotting (data not shown). In contrast with its effect on JNK, the Dbl expression vector did not stimulate HA-ERK2 activity (Figure 6B).

Activation of JNKK by Rac1

The activation of both JNK1 and p38 by Rac1 suggested that the protein kinase that can activate both of these MAPKs, SEK1/MKK4/JNKK, may mediate this response. Indeed, we found that cotransfection with the Rac1(V12) vector activated a GST-JNKK chimera whose activity was measured by a coupled kinase assay (Figure 7A). While the response to Rac1 is weaker than the response to MEKK1, it is worth noting that MEKK1 is a direct JNKK activator whose expression appears to be rate limiting, as even a small increase in its expression results in consider-

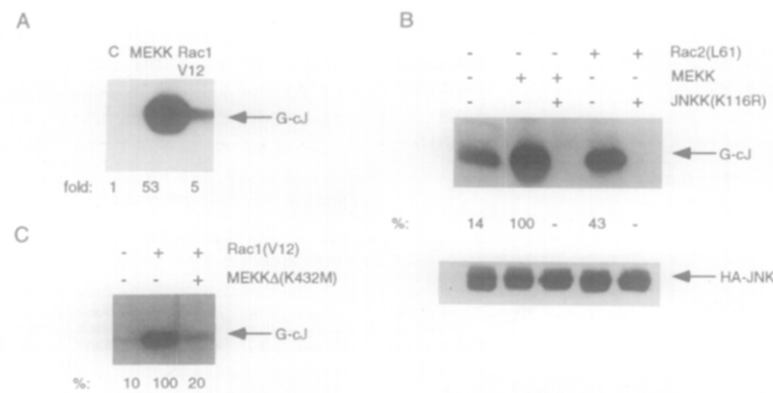


Figure 7. JNKK and MEKK Act Downstream of Rac

(A) COS-1 cells were transfected with a mammalian GST-JNKK expression vector together with empty expression vector (lane C) or with the indicated expression vectors for MEKK and Rac1(V12). GST-JNKK was isolated from cell lysates by binding to glutathione-Sepharose and JNKK activity was determined by a coupled kinase assay using recombinant JNK1 and GST-c-Jun (G-cj) as substrates. This experiment was repeated twice with similar results.

(B) COS-1 cells were transfected with pSRα-HA-JNK1, pCMV5-MEKK, pSRα-Rac2(L61), pSRα-JNKK(K116R), or empty expression vectors, as indicated. JNK activity and expression

were determined as described above. The levels of JNK activity are expressed relative to those seen in cells cotransfected with MEKK. Owing to higher basal levels in COS-1 cells, the fold stimulation is not as high as in HeLa or NIH 3T3 cells.

(C) HeLa cells were cotransfected with HA-JNK1 and either Rac1(V12), MEKKΔ(K432M), or empty expression vectors, as indicated. HA-JNK1 activity was measured as described above and is expressed relative to the level found in Rac1(V12)-transfected cells. This experiment was repeated twice with similar results.

able JNKK or JNK activation (Lin et al., 1995; Minden et al., 1994). Consistent with these results, we found that cotransfection with a vector encoding catalytically inactive JNKK blocked JNK activation by Rac2(L61) (Figure 7B). JNK activation by Rac1(V12) was also inhibited by cotransfection of a plasmid encoding catalytically inactive MEKK1 (Figure 7C). Rac1(N17) did not inhibit JNK activation by MEKK1 (data not shown).

Stimulation of c-Jun Transcriptional Activity by Rac and Cdc42Hs

The efficient activation of the JNK cascade by Rac1/Rac2 and Cdc42Hs should result in stimulation of c-Jun transcriptional activity. Indeed, both Rac2(L61) and Cdc42Hs(V12) stimulated the transcriptional activity of a GAL4-c-Jun (1-223) fusion protein containing the c-Jun activation domain (Figure 8). This stimulation is likely to depend on c-Jun phosphorylation because neither Rac1(V12) nor Cdc42Hs(V12) stimulated the activity of a GAL4-c-Jun (1-223;A63/73) fusion protein in which the serines that are phosphorylated by JNK were replaced by alanines.

Discussion

Members of the Rho subfamily of small GTPases, Rac1, RhoA, and Cdc42Hs, mediate effects of growth factors and polypeptide hormones on the actin cytoskeleton and thereby affect cell morphology (Ridley and Hall, 1992; Ridley et al., 1992; Kozma et al., 1995; Nobes and Hall, 1995). These functions are evolutionary conserved, as the Cdc42Sc protein is involved in bud site selection and polarized cell growth (Adams et al., 1990). There is also circumstantial evidence that the Rho-like proteins may have additional effects on the control of cell growth and proliferation. For example, several members of the Dbl family, which may function as nucleotide exchange factors for the Rho-like proteins, have oncogenic activity (Adams et al., 1992; Hart et al., 1991, 1994; Miki et al., 1993). Furthermore, activated Rac1 cooperates with a membrane-targeted

form of Raf (Raf-CAAX) in oncogenic transformation (Qiu et al., 1995). Although these effects could be mediated through modulation of the cytoskeleton, since oncogenic transformation and Ha-Ras activation in particular are associated with cell shape changes (Bar-Sagi and Feramisco, 1986; Ridley, 1994), we show that the functions of Rac1/Rac2 and Cdc42Hs are not limited to their effects on cell morphology. These proteins are efficient activators of a signaling cascade affecting two subgroups of MAPKs: the JNKs/SAPKs and p38/Mpk2. Interestingly, another member of this subfamily, RhoA, is incapable of activating this cascade, and none of the Rho-like proteins examined affected the ERK subgroup of MAPKs. Consistent with activation of the JNK cascade, both Rac1 and Cdc42Hs stimulated the transcriptional activity of c-Jun. As the JNKs and probably p38 are likely to be capable of phosphorylating other transcription factors, as well as regulatory proteins that act posttranscriptionally (Lee et al., 1994), Rac1/Rac2 and Cdc42Hs may mediate effects of growth factors and polypeptide hormones on gene expression. In that respect, they are similar to members of the Ras subfamily of GTPases, whose function in linking growth factor receptor activation to transcriptional responses via MAPK activation is well established (Egan and Weinberg, 1993; Marshall, 1995; Hill and Treisman, 1995).

The role of Rac1/Rac2 and Cdc42Hs in JNK activation was established by three types of experiments. First, we demonstrated that constitutively activated Rac1(V12), Rac2(L61), and Cdc42Hs(V12) stimulate the catalytic activities of JNK1 and p38 by 10- to 60-fold in transient transfection assays in two different cell types, even when expressed at relatively low levels. Stable expression of Rac1(V12) or Cdc42Hs(V12) in a third cell type also results in efficient JNK activation, albeit of a lower magnitude than the one achieved in transient transfection experiments. This difference is not surprising, as stably transfected cells are likely to activate feedback pathways as part of their adaptation to chronic stimulation. In addition to JNK and p38 activation, we demonstrated that Rac1 activates

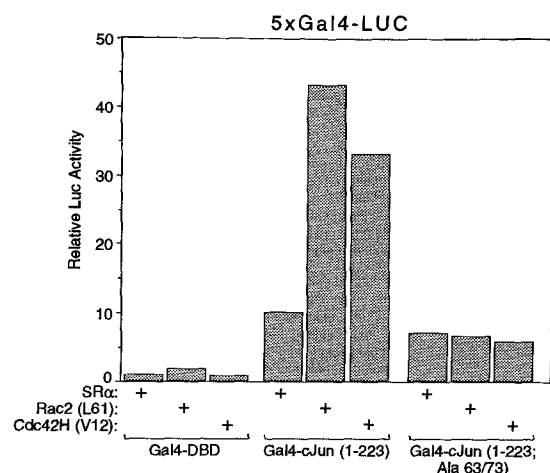


Figure 8. Rac2 and Cdc42Hs Potentiate c-Jun-Activated Transcription
CV1 cells were transiently cotransfected with 1 μ g of a 5 $\times$ Gal4-LUC reporter and 10 ng of expression vectors encoding either the GAL4 DNA-binding domain (DBD), GAL4-c-Jun(1-223), or GAL4-c-Jun(1-223;A63/73) fusion proteins, and 20 ng of either Rac2(L61), Cdc42Hs(V12), or the empty pSR α vector. Luciferase activity was determined 16–20 hr after transfection. The results are expressed relative to the level of luciferase activity in cells cotransfected with GAL4-DBD and pSR α , which was given an arbitrary value of 1. The average of three experiments done in duplicate are shown.

JNKK, a MAPKK that activates both of these MAPK types. These effects are specific, as ERK activity was not considerably stimulated even after transfection of rather high levels of Rac1/Rac2 or Cdc42Hs expression vectors. Second, we used a dominant interfering Rac1 allele to block JNK activation by Ha-Ras, v-Src, or EGF. Based on the Ha-Ras(N17) dominant negative mutant (Feig and Cooper, 1988), Rac1(N17) is expected preferentially to bind GDP versus GTP and thereby have a higher relative affinity toward its exchange factor than its wild-type counterpart. The inhibitory effect of Rac1(N17) is specific, as it did not have a considerable effect on JNK activation by either TNF α , which occurs through a Ras-independent mechanism (Minden et al., 1994), or by constitutively activated Rac1(V12), which no longer requires an exchange factor. In addition, it does not inhibit JNK activation by a downstream component, MEKK1 (data not shown). The third type of experiment shows that expression of the Dbl oncoprotein, which can act as an exchange factor for Cdc42Hs (Hart et al., 1991, 1994), also stimulates JNK activity. This effect of Dbl is inhibited by Rac1(N17) but not by Ha-Ras(N17). JNK activation by Dbl was also blocked by expression of the regulatory domain of PAK65, which binds to the GTP-loaded forms of Rac1/Rac2 and Cdc42Hs, but not RhoA (Manser et al., 1994; Martin et al., 1995). As RhoA is not involved in JNK activation and Dbl does not act on Rac, these results suggest that Cdc42Hs may represent a physiological target for Dbl.

The use of the dominant interfering Rac1(N17) mutant allowed us to place Rac1 as a critical intermediate between Ha-Ras and MEKK1 in the signaling pathway leading from activated receptor and nonreceptor tyrosine

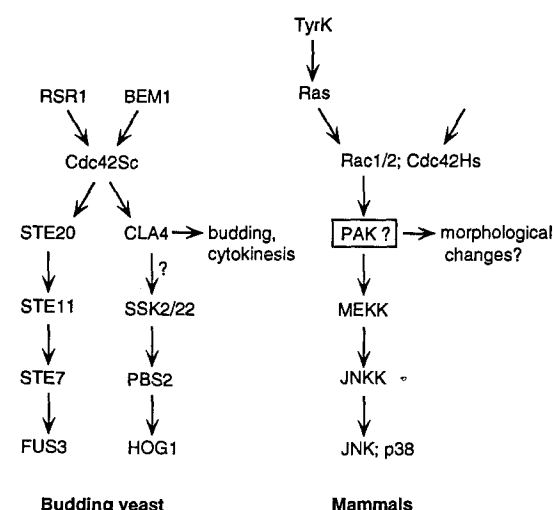


Figure 9. Signal Transduction through Members of the Rho Subfamily of Small GTPases in Budding Yeast and Mammals

The regulation of yeast and mammalian Rac1/Rac2 and Cdc42 and their downstream effects on cell morphology and MAPK cascades are summarized. Cdc42Sc is activated (through Cdc24; not shown) by RSR1, a Ras protein, and by BEM1, an SH3 domain protein (Bender and Pringle, 1989; Chenevert et al., 1992; Chang et al., 1994). Rac1/Rac2 and Cdc42Hs also receive input from growth factor receptors and possibly trimeric G protein-coupled receptors through mediators yet to be identified (Kozma et al., 1995; Nobes and Hall, 1995). Activated Cdc42Sc can participate in activation of STE20 (I. Herskowitz, personal communication) and CLA4 (Cvrčková et al., 1995), thereby affecting the pheromone response and osmotic shock MAPK cascades, respectively. Cdc42Sc induces morphological changes associated with budding and is required for cytokinesis, possibly through CLA4 (Cvrčková et al., 1995). Activated Rac1/Rac2 and Cdc42Hs induce a variety of morphological changes through the actin cytoskeleton (Kozma et al., 1995; Nobes and Hall, 1995; Ridley, 1994). In addition, they probably activate PAK family members. PAK activation by analogy to CLA4 could mediate some of the morphological changes and, in addition, is speculated to activate the JNK and p38 cascades.

kinases to JNK. Although MEKK1 activity has been suggested to be stimulated by Ha-Ras (Lange-Carter and Johnson, 1994), the mechanism of its activation is unknown. Unlike its fission yeast homolog Byr2, MEKK1 does not interact directly with Ha-Ras (Chang et al., 1994; White et al., 1995).

In trying to sort out the epistatic relationships between Rac1/Rac2 and Cdc42Hs, we also used a dominant interfering Cdc42Hs(N17) mutant (Kozma et al., 1995). However, the results obtained did not allow us to draw clear conclusions. While Rac1(N17) inhibited JNK activation by Cdc42Hs(V12), partial but considerable inhibition of JNK activation by Rac2(L61) was obtained by expression of Cdc42Hs(N17). In addition, this mutant may have some nonspecific effect, as it also partially inhibited JNK activation by Cdc42Hs(V12) (data not shown). Thus, while it is plausible that Cdc42Hs acts upstream of Rac1 and Rac2, as previously suggested based on their morphological effects (Kozma et al., 1995; Nobes and Hall, 1995), it is worth noting that the effect of Cdc42Hs(N17) on Rac-induced lamellipodia formation was not examined. Rac1/Rac2 and Cdc42Hs may interact with a common exchange factor,

thus explaining the cross inhibition. However, the exact identity of the relevant exchange factors and the biochemical details of the interactions among the Rho proteins are not clear (Nobes and Hall, 1994). Furthermore, certain proteins like Bcr contain a Dbl domain and a GTPase-activating domain for both Rac and Cdc42Hs (Diekmann et al., 1991). Thus, although Cdc42Hs probably acts upstream of Rac, the Cdc42Hs(N17) mutant may titrate a protein that is an important Rac effector.

Although Rac1 acts downstream of Ha-Ras in the pathway leading to either lamellipodia formation (Nobes and Hall, 1995; Ridley et al., 1992) or JNK activation, it is not known how Ras activates Rac (Nobes and Hall, 1994; Ridley, 1994). Whereas Ha-Ras(N17) completely blocks JNK activation by EGF or v-Src, Rac1(N17) leads to only partial inhibition. Although the basis for these differences could be trivial, as Rac1(N17) is not efficiently expressed (our unpublished data), we note that even higher doses of Rac1(N17) did not completely inhibit JNK activation by the tyrosine kinases. A more likely explanation for these and previous observations, that Ha-Ras is necessary but not sufficient for JNK activation in response to EGF or nerve growth factor (Minden et al., 1994), is that receptor activation generates another signal that synergizes with Ras to activate the JNK cascade. In T cells, for example, activation of Ras either through TCR occupancy or phorbol esters results in modest JNK activation that is strongly potentiated following treatment with Ca^{2+} ionophore or activation of the CD28 auxiliary receptor (Su et al., 1994). By itself, Ca^{2+} ionophore or CD28 does not stimulate JNK activity. Thus, the residual level of Ras activity following Rac1(N17) expression may be sufficient to synergize with the second signal to increase JNK activity. As this synergizing signal is not produced by Ha-Ras(L61), the inhibition of its activity by Rac1(N17) is more extensive. These results also suggest that the relationships among tyrosine kinases, Ha-Ras, and Rac (and Cdc42Hs) may be more complex than a single sequential cascade. A somewhat similar situation exists in *Schizosaccharomyces pombe*, where the Ha-Ras homolog Ras1 activates Scd1, a homolog of Cdc24Sc, the exchange factor for Cdc42Sc (Chang et al., 1994). Both Cdc24 and Scd1 contain a Dbl domain and are activated by BEM1 and Scd2, respectively (Cheney et al., 1992; Chang et al., 1994). BEM1 and Scd2 are homologous proteins that contain SH3 domains. In Cdc42Sc, activity is also affected by RSR1/BUD1, a Ras-like protein involved in bud site selection (Bender and Pringle, 1989) that interacts with Cdc24 (I. Herskowitz, personal communication). Thus, as illustrated in Figure 9, both mammalian and yeast members of the Rho subfamily receive input from Ras-like proteins and other sources. Mammalian Cdc42 was suggested to be activated by bradykinin, which operates through a trimeric G protein-coupled receptor (Kozma et al., 1995), and Rac was suggested to be activated by phosphatidylinositol 3-kinase in addition to Ha-Ras (Nobes and Hall, 1995).

This similarity extends to the downstream functions of the Rho-like proteins. Mutant alleles of Ras1, Scd1, and Scd2 affect both mating and cell morphology (Chang et al., 1994). While Ras1 also activates both Byr2 and the MAPKKK that acts in the pheromone signaling cascade

(Herskowitz, 1995), it was not clear how Scd1 and Scd2 (Cdc24 and Cdc42) affect mating. It was recently found that Cdc42Sc is required for activation of the pheromone cascade, most likely through direct binding to STE20 (I. Herskowitz, personal communication). In vitro, Cdc42·GTP stimulates STE20 autocatalytic activity (A. A., unpublished data). Another potential connection between Cdc42 and MAPK activation is provided by CLA4, a novel STE20-like kinase required for proper budding and cytokinesis in *S. cerevisiae*, but not for the pheromone-responsive MAPK cascade (Cvrčková et al., 1995). The Cla4 phenotype is rescued by growth on high salt in a manner that depends on activation of the osmotic stress MAPK module, which includes HOG1 and PBS2, suggesting a relationship between them (Cvrčková et al., 1995). Recently, two MEKK-like MAPKKs, SSK2 and SSK22, that activate PBS2 were identified (Maeda et al., 1995). Both JNK1 and p38 partially complement *hog1* mutants (Galcheva-Gargova et al., 1994; Han et al., 1994), and JNKK complements *pbs2* mutants (Lin et al., 1995). Thus, these results outline a remarkable evolutionary conservation in regulation of the osmotic stress and budding pathways in yeast and JNK and the cell morphology pathways in mammals (Figure 9).

Once activated, Rac1, Cdc42Sc, and Cdc42Hs bind and presumably activate STE20-like kinases. In yeast, Cdc42Sc binds and presumably activates both CLA4 (Cvrčková et al., 1995) and STE20 (I. Herskowitz, personal communication). The identity of the STE20-like kinase acting in the mammalian pathway is currently unknown. Although we were able to inhibit JNK activation by expressing the regulatory domain of PAK65, we have not succeeded in constructing a constitutively activated PAK65 allele to obtain positive evidence for its involvement in this pathway. In light of the results described above, the inhibitory effect of the PAK65 regulatory domain activation is not surprising, as it is expected to titrate both Rac1 and Cdc42Hs and therefore it does not constitute evidence for the involvement of PAK65 or other family members (M. Cobb, personal communication) in JNK activation. However, based on the yeast system, we proposed that a STE20/PAK homolog acts between Rac and MEKK.

At this point it is not clear whether the JNK and p38 cascades are involved in any of the morphological changes seen in response to Rac or Cdc42Hs activation. Most likely, JNK and p38 do not directly affect these changes because the morphological responses to Rac and Cdc42Hs are distinct while their effect on the MAPK cascades are very similar. However, it is likely that JNK and possibly p38 mediate some of the long-term effects of Rac1, Cdc42Hs, and Dbl on cell growth, differentiation, and proliferation. The involvement of JNK in stimulation of AP-1 activity suggests it may be responsible for the increased invasiveness of Rac1(V12)-expressing cells (Michiels et al., 1995) through induction of metalloproteinase expression.

Experimental Procedures

Cell Culture and Transfections

NIH 3T3, COS-1, HeLa, and CV1 cells were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bo-

vine serum (FBS), 1 mM glutamate, 100 U/ml penicillin, and 100 µg/ml streptomycin. For transient transfection assays, HeLa, COS-1, and NIH 3T3 cells grown in 35 mm dishes were transfected using the lipofectamine (GIBCO BRL) method according to the protocol of the manufacturer, keeping total DNA at 2 µg per plate. Cells were harvested after 48 hr. CV1 cells were transfected by calcium phosphate coprecipitation, keeping total DNA at 5 µg per plate. Luciferase activity was assayed as described elsewhere (Lin et al., 1995). For transformation assays, NIH 3T3 cells were cultured in DMEM supplemented with 10% donor calf serum (DCS). Cells in 100 mm dishes were transfected with plasmid DNA together with 20 µg of human placental DNA (a gift from M. Symons) by calcium phosphate coprecipitation. After 24 hr, cells were kept in medium containing 5% DCS. The medium was changed every 4 days, and the number of foci was determined 10–14 days later. To generate stably transfected cell lines, we transfected Rat1 fibroblasts with pCVN expression vectors containing either Rac1(V12) or Cdc42Hs(V12) alleles. Prior to transfections, cells were trypsinized, washed, and resuspended at 1.5×10^6 cells/ml in cytomix buffer containing 10 mM K_2HPO_4/KH_2PO_4 (pH 7.5), 25 mM HEPES (pH 7.6), 2 mM EGTA, 5 mM $MgCl_2$, 2 mM ATP, and 5 mM glutathione. The cell suspensions were mixed with 10 µg of plasmid and salmon testis DNA and electroporated at 125 µF, 250 V. Cells were then plated in 100 mm dishes, grown for 16 hr, trypsinized, resuspended in 5 ml of high glucose DMEM supplemented with 10% FBS, 2 mM glutamine, 100 µg/ml streptomycin, 100 U of penicillin, and plated at various dilutions in medium containing G418 (400 µg/ml). G418-resistant clones were selected, and positive clones were identified by immunostaining as described elsewhere (Martin et al., 1995). Positive clones were grown in 100 mm dishes to 70%–80% confluency, followed by serum starvation for 16 hr prior to harvesting and assaying endogenous JNK and ERK activities.

Protein Kinase Assays

To assay HA-JNK1 and HA-ERK2 activity, we collected cells from 35 mm plates transfected with the various vectors in 400 µl of lysis buffer 48 hr after transfection, as described elsewhere (Minden et al., 1994). The kinases were immunoprecipitated with anti-HA antibody, and their activities were measured using 2 µg of either GST-c-Jun(1–79) or MBP as substrates (Minden et al., 1994; Dérjard et al., 1994). M2-p38 activity was similarly assayed after immunoprecipitation with the anti-Flag antibody (IBI/Kodak) using 2 µg of GST-ATF2(1–254) (GST-ATF2Δ) as a substrate (Dérjard et al., 1995; Lin et al., 1995). Endogenous JNK and ERK activities were also assayed by immunocomplex kinase assays after immunoprecipitation of JNK1/JNK2 with anti-JNK(247) and anti-ERK2 TR10 antibodies as described elsewhere (Minden et al., 1994). TR10 was a gift from M. Weber. All experiments were quantitated using a phosphorimager.

Immunoblotting

Immunocomplexes or 25 µg of whole-cell lysates were resolved by SDS-PAGE and electroblotted onto Immobilon P membranes (Millipore). After blocking, the membranes were probed with anti-HA antibody or anti-JNK monoclonal antibody 333.8 (Pharmingen) as described elsewhere (Minden et al., 1994). Immunocomplexes were visualized using enhanced chemiluminescence detection (Amersham International).

Plasmids

Rac1(N17), RhoA(V14), and RhoA(N19) were expressed in the pEXV vector (Ridley and Hall, 1992; Ridley et al., 1992). Rac1(V12) (Ridley et al., 1992) was subcloned into pSRα (Takabe et al., 1988) at the EcoRI site. Rac2(L61) (Xu et al., 1994) was subcloned into pSRα between the EcoRI and HindIII sites. Cdc42Hs(V12) (Kozma et al., 1995) and Dbl (Hart et al., 1991) were expressed using pCMV and pEXV expression vectors, respectively. pSG-v-Src was a gift from T. Deng. The HA-JNK1, HA-ERK2, M2-p38, MEKK1, MEKKΔ(K432M), human JNKK, human JNKK(K116R), JNK1(APF), GAL4-c-Jun(1–223), and GAL4-c-Jun(1–223;Δ63/73) expression vectors were described (Dérjard et al., 1994; Su et al., 1994; Minden et al., 1994; Lin et al., 1995). GST-ATF2(1–254) (GST-ATF2Δ) was constructed by inserting a BamHI–NcoI fragment of pRSET-His ATF2 (Abdel-Malek Abdel Hafiz et al., 1992) into pGEX-KG, a GST fusion protein expression vector. MKK3 expression vector was a gift from K. Guan. PAKR contains the N-term

inal Rac-binding domain of human PAK65 (amino acids 1–225) (Martin et al., 1995) and was expressed using the pCDB vector.

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